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CPE 512 Project Proposal - Parallel N-Body Gravitational Simulation

The N-body problem simulates how multiple particles move under mutual gravitational forces. It is a core technique in astrophysics for modeling everything from planetary systems to star clusters. Because calculating every pairwise interaction requires $O(N^2)$ work per timestep, brute-force simulations quickly become computationally bottlenecked as particle counts grow. This project explores how shared-memory parallel computing can alleviate that bottleneck and significantly accelerate execution times.

Using C++ and OpenMP on the Alabama Supercomputer Authority's ASA-X system, I will build a parallel direct-force N-body simulation. The pipeline updates particle forces, velocities, and positions across discrete timesteps. To establish a baseline for both performance and output verification, I will first develop a serial implementation. The parallel version will then partition the pairwise gravitational force calculations across available CPU threads, taking care to manage thread work distribution and minimize synchronization overhead.

The core objective is to evaluate how effectively OpenMP scales as both particle counts and thread counts increase. Performance will be measured using total execution time, speedup and parallel efficiency. Correctness will be verified by comparing parallel outputs against the serial reference baseline.

Ultimately, this project will deliver a functional parallel N-body solver along with a clear analysis of its scaling characteristics on the ASA-X hardware. The results will highlight how problem size influences parallel efficiency and pinpoint whether performance ceilings stem from memory bandwidth limitations, thread overhead, or hardware constraints. All work will leverage the standard C++ compilers and OpenMP runtime already configured on the ASA-X system, requiring no specialized hardware or third-party software.